

Product info

KonstruX ST ZK 6,5x140

Article number 904811

100 Piece

Length

140 mm

Length screw head

5.5 mm

Number of screws per joint 2

Diameter screw / thread

6.5 mm

Diameter head

8.0 mm

Diameter core

4.5 mm

User input

Side flaps

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled

Width = 100 mm | Height = 120 mm | Overlap = 250 mm

Carrier

Pinewood | C24 | Spruce, pine, fir | Not pre-drilled

Width = 160 mm | Height = 240 mm

Load

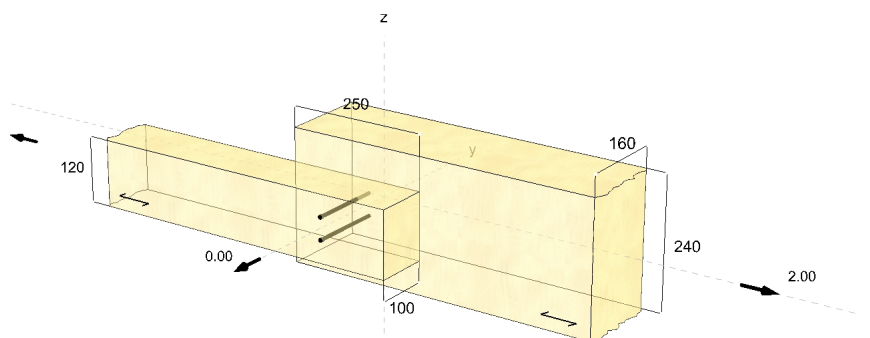
Class of usage: 1

Design loads

x-direction = 2.00 kN | Duration of load: medium

Screws

Configuration: straight | Screwing: flush beam

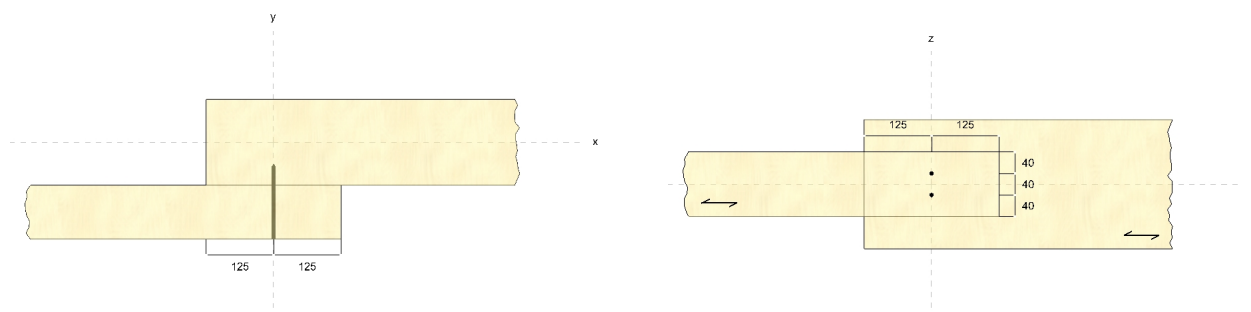


Screw distances

Screwing angle	90 °
Drilling depth	0 mm

Distances - Side flaps [mm]	Minimum	Existing	
$a_{3,c}$	98	125	EN 1995-1-1
$a_{4,t}$	33	40	EN 1995-1-1
a_2	33	40	EN 1995-1-1

Distances - Carrier [mm]	Minimum	Existing	
$a_{3,c}$	98	125	EN 1995-1-1
$a_{4,t}$	33	100	EN 1995-1-1
a_2	33	40	EN 1995-1-1



Design

Design loads

$$V_{d,S} = F_{v,d} = 2.00 \text{ kN}$$

Stress in x-direction

Thread in the side flaps

$$n = 1$$

$$n_{ef} = n^{0.9} = 1.00$$

$$\alpha = 90^\circ$$

$$k_{ax} = 1.0$$

$$f_{ax,k} = 11.40 \frac{N}{mm^2}$$

$$d = 6.5 mm$$

$$l_{ef} = 95 mm$$

$$\rho_k = 350 \frac{kg}{m^3}$$

$$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350} \right)^{0.8} = 7.00 kN$$

EN 1995-1-1
8.7.2 (8) (8.41)

ETA-11/0024

ETA-11/0024

ETA-11/0024

ETA-11/0024

EN 338 5
EN 14080 5.1.4.3 (4)(5)

ETA-11/0024

Thread in the support

$$n = 1$$

$$n_{ef} = n^{0.9} = 1.00$$

$$\alpha = 90^\circ$$

$$k_{ax} = 1.0$$

$$f_{ax,k} = 11.40 \frac{N}{mm^2}$$

$$d = 6.5 mm$$

$$l_{ef} = 36 mm$$

$$\rho_k = 350 \frac{kg}{m^3}$$

$$F_{ax,\alpha,Rk} = n_{ef} \cdot k_{ax} \cdot f_{ax,k} \cdot d \cdot l_{ef} \cdot \left(\frac{\rho_k}{350} \right)^{0.8} = 2.65 kN$$

EN 1995-1-1
8.7.2 (8) (8.41)

ETA-11/0024

ETA-11/0024

ETA-11/0024

ETA-11/0024

EN 338 5
EN 14080 5.1.4.3 (4)(5)

ETA-11/0024

Uses formula for nails (8.23, 8.24)

Why not screws (8.38), 8.40a ?

Tensile load capacity

$$n = 1$$

$$n_{ef} = n^{0.9} = 1.00$$

$$f_{tens,k} = 17.00 kN$$

$$F_{t,Rk} = n_{ef} \cdot f_{tens,k} = 17.00 kN$$

EN 1995-1-1
8.7.2 (8) (8.41)

ETA-11/0024

EN 1995-1-1
8.7.2 (7) (8.40c)

Shear

$$V_{d,S} = 2.00 \text{ kN}$$

$$k_{mod,1} = 0.80$$

EN 1995-1-1
3.1.3 (1)

$$k_{mod,2} = 0.80$$

EN 1995-1-1
3.1.3 (1)

$$k_{mod} = \sqrt{k_{mod,1} k_{mod,2}} = 0.80$$

EN 1995-1-1
2.3.3.1 (2) (2.6)

$$n_{0,1} = 1$$

$$n_{ef,0,1} = 1.00$$

EN 1995-1-1
8.3.1.1 (8) (8.17)

$$n_{90,1} = 2$$

$$n_{0,2} = 1$$

$$n_{ef,0,2} = 1.00$$

EN 1995-1-1
8.3.1.1 (8) (8.17)

$$n_{90,2} = 2$$

$$n = \min(n_{ef,0,1} \cdot n_{90,1}; n_{ef,0,2} \cdot n_{90,2}) = 2.00$$

$$\rho_{k,1} = 350 \frac{\text{kg}}{\text{m}^3}$$

EN 338 5
EN 14080 5.1.4.3 (4)(5)

$$\rho_{k,2} = 350 \frac{\text{kg}}{\text{m}^3}$$

EN 338 5
EN 14080 5.1.4.3 (4)(5)

$$f_{h,1,k} = 16.37 \frac{\text{N}}{\text{mm}^2}$$

ETA-11/0024

$$f_{h,2,k} = 16.37 \frac{\text{N}}{\text{mm}^2}$$

ETA-11/0024

$$t_1 = 100 \text{ mm}$$

EN 1995-1-1
8.2.2 (1)

$$t_2 = 40 \text{ mm}$$

EN 1995-1-1
8.2.2 (1)

$$\beta = \frac{f_{h,2,k}}{f_{h,1,k}} = 1.00$$

EN 1995-1-1
8.2.2 (1) (8.8)

$$M_{y,k} = 15.0 \text{ Nm}$$

ETA-11/0024

$$F_{ax,Rk} = 2.65 \text{ kN}$$

EN 1995-1-1
8.2.2 (2)

$$c) f_{h,1,k} t_1 d = 10.64 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$b) f_{h,2,k} t_2 d = 4.26 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$c) \frac{f_{h,1,k} t_1 d}{1+\beta} \left[\sqrt{\beta + 2\beta^2 \left[1 + \frac{t_2}{t_1} + \left(\frac{t_2}{t_1} \right)^2 \right] + \beta^3 \left(\frac{t_2}{t_1} \right)^2} - \beta \left(1 + \frac{t_2}{t_1} \right) \right] + \frac{F_{ax,Rk}}{4} = 4.22 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$d) 1.05 \frac{f_{h,1,k} t_1 d}{2+\beta} \left[\sqrt{2\beta(1+\beta) + \frac{4\beta(2+\beta)M_{y,k}}{f_{h,1,k} d t_1^2}} - \beta \right] + \frac{F_{ax,Rk}}{4} = 4.54 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$e) 1.05 \frac{f_{h,1,k} t_2 d}{1+2\beta} \left[\sqrt{2\beta^2(1+\beta) + \frac{4\beta(1+2\beta)M_{y,k}}{f_{h,1,k} d t_2^2}} - \beta \right] + \frac{F_{ax,Rk}}{4} = 2.52 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$f) 1.15 \sqrt{\frac{2\beta}{1+\beta}} \sqrt{2M_{y,k} f_{h,1,k} d} + \frac{F_{ax,Rk}}{4} = 2.72 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$F_{v,Rk} = \min(F_{v,Rk}(c); F_{v,Rk}(b); F_{v,Rk}(c); F_{v,Rk}(d); F_{v,Rk}(e); F_{v,Rk}(f)) = 2.52 \text{ kN}$$

EN 1995-1-1
8.2.2 (1) (8.6)

$$\gamma_M = 1.30$$

ÖNORM B 1995-1-1
2.4.1(1) P



$$F_{v,Rd} = k_{mod} \cdot \frac{F_{v,Rk}}{\gamma_M} = 1.55 \text{ kN}$$

EN 1995-1-1
2.4.3 (1)P (2.17)

$$\eta = \left(\frac{V_{dS}}{n \cdot F_{v,Rd}} \right) \cdot 100 \% = 64.42 \%$$

The system has been calculated successfully

Hints

- The calculation is according to:
- EN 338 (2016-07) + EN 14081-1 (2016-06) EN 14080 (2013-09)
- EN 1990 (2010-12)
- EN 1991-1-1 (2010-12)
- EN 1993-1-1 (2010-12)
- EN 1995-1-1 (2010-12) + EN 1995-1-1/A2 (2014-07) + ÖNORM B 1995-1-1 (2015-06)
- DIN EN 1995-1-1/NA (2013-08)
- ETA-11/0024 (2017-03)
- The screws may only be used when subjected predominantly to dead loads.
- Verification of the components must be carried out separately if necessary.
- Attention: Please check the assumptions. The displayed values, types and numbers of fasteners are dimensioning aids only.
Projects can only be dimensioned by authorized people.
If you need a verification of stability please contact an authorized civil engineer. We can gladly help you with respective contacts.